Lipid Digestion Absorption and Transport

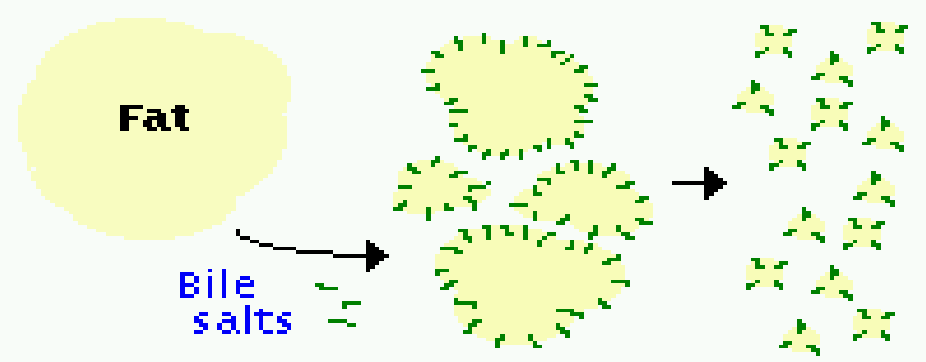
prepared by
Fatima Noor

Monogastric Digestion

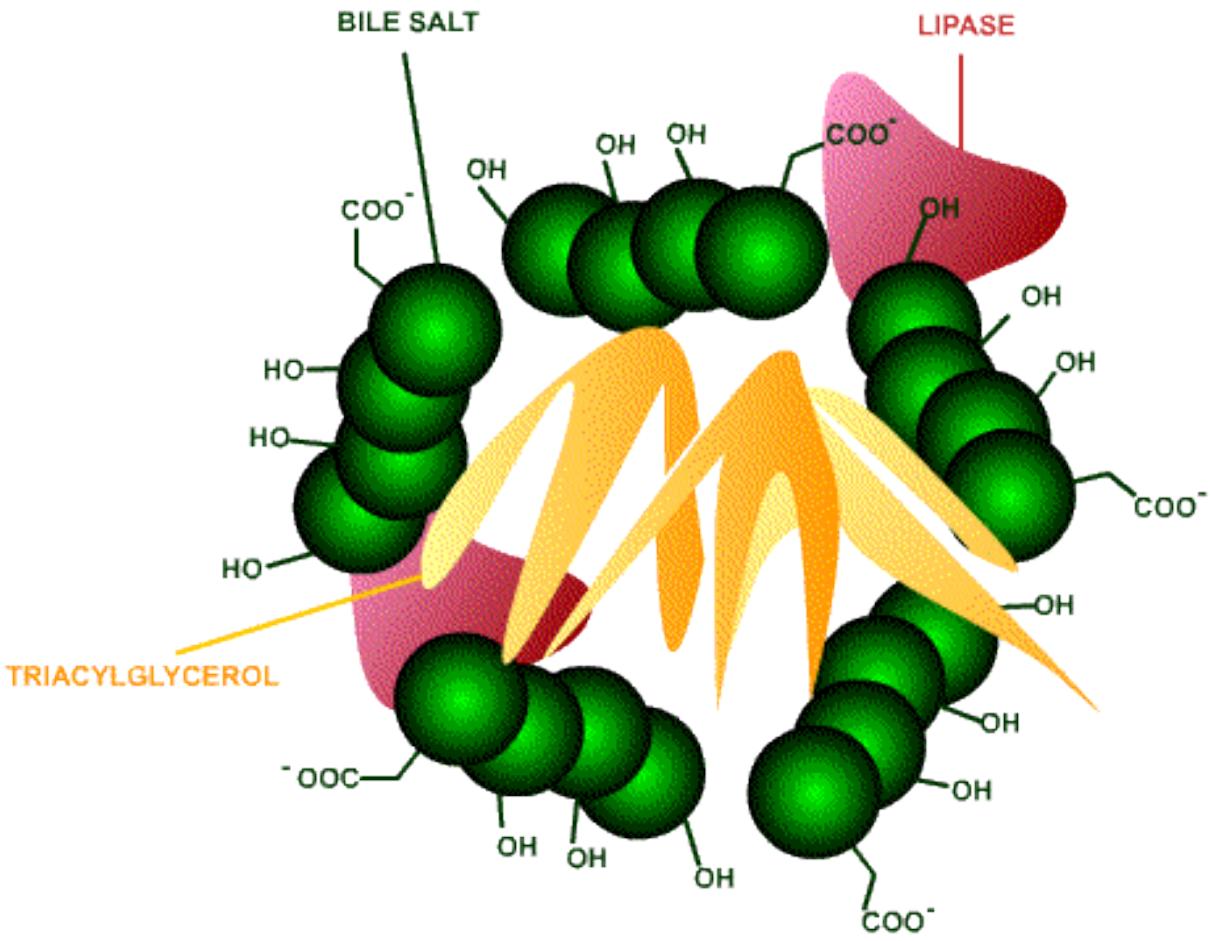
- Challenges
  - Lipids are not water soluble
  - Triglycerides too large to be absorbed
- Digestive solution
  - Triglycerides mix with bile and pancreatic secretions
    - Emulsification and digestion

Bile

- Produced in liver, stored in gallbladder
  - Except horse
- Alkaline solution composed of:
  - Bile salts
  - Cholesterol
  - Lecithin
  - Bilirubin



- Responsible for fat emulsification
  - Detergent action



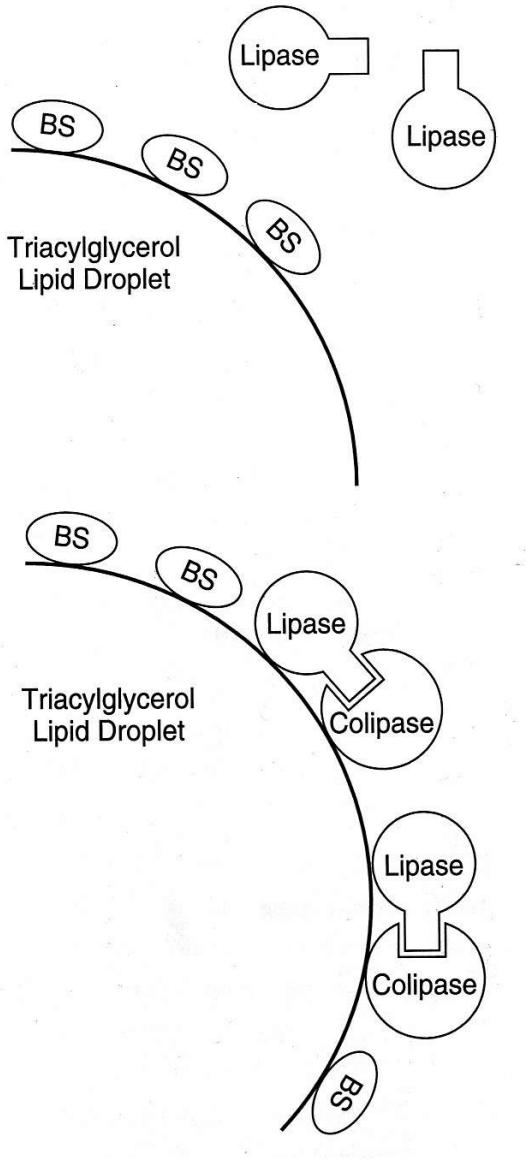
Mixed micelle formed by bile salts, triacylglycerols and pancreatic lipase.

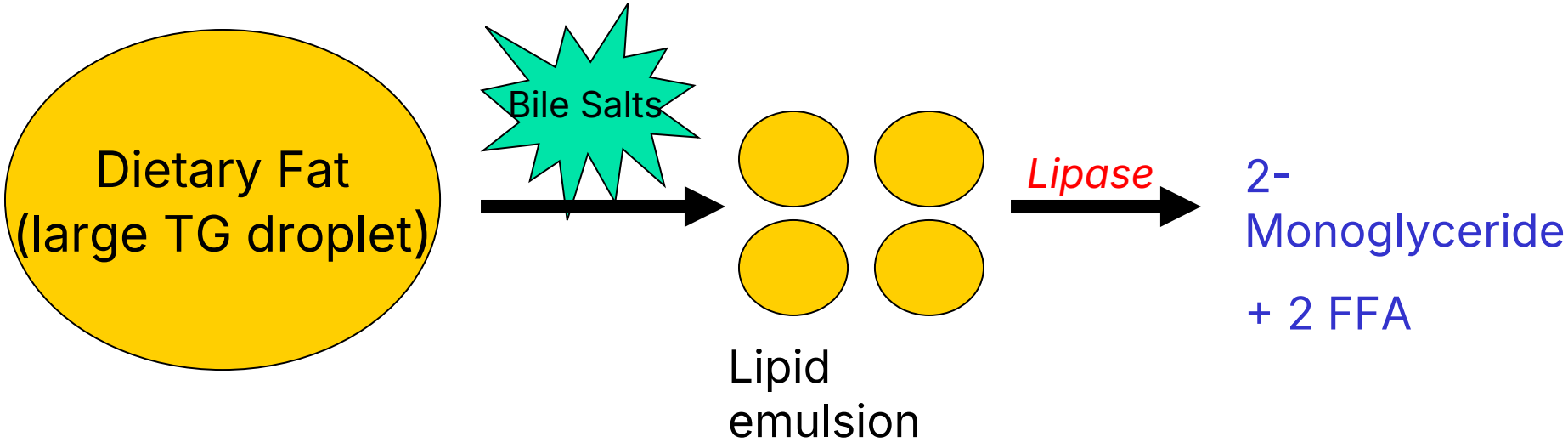
Digestion of Lipid

- Bile salts emulsify lipids
- Pancreatic lipase acts on triglycerides
  - Triglycerides ⟶ *sn*-2 monoglyceride + 2 fatty acids
- Pancreatic colipase
  - Activated by trypsin
  - Interacts with triglyceride and pancreatic lipase
    - Displaces bile to allow recycling
    - Improves activity of pancreatic lipase

Pancreatic Colipase

- Secreted from pancreas as procolipase
  - Activated (cleaved) by trypsin
- Anchors lipase to the micelle
  - One colipase to one lipase (i.e., 1:1 ratio)




Dietary Fat
(large TG droplet)
Bile Salts
Lipid
emulsion
Lipase
2-
Monoglyceride
+ 2 FFA

Digestion of Lipid

- Phospholipase A_1 and A_2
  - Hydrolyzes fatty acids from phospholipids
- Cholesterol esterase
  - Hydrolyzes fatty acids from cholesterol esters

Micelle Formation

- Complex of lipid materials soluble in water
- Contains bile salts, phospholipids & cholesterol
- Combines with 2-monoglycerides, free fatty acids and fat-soluble vitamins to form mixed micelles

Lipid Absorption

- Mixed micelles move to intestinal mucosal cells (enterocytes) and release contents near cell
- The bile salts are re-absorbed further down the gastrointestinal tract (in the ileum), transported to the liver, and finally recycled and secreted back into the digestive tract

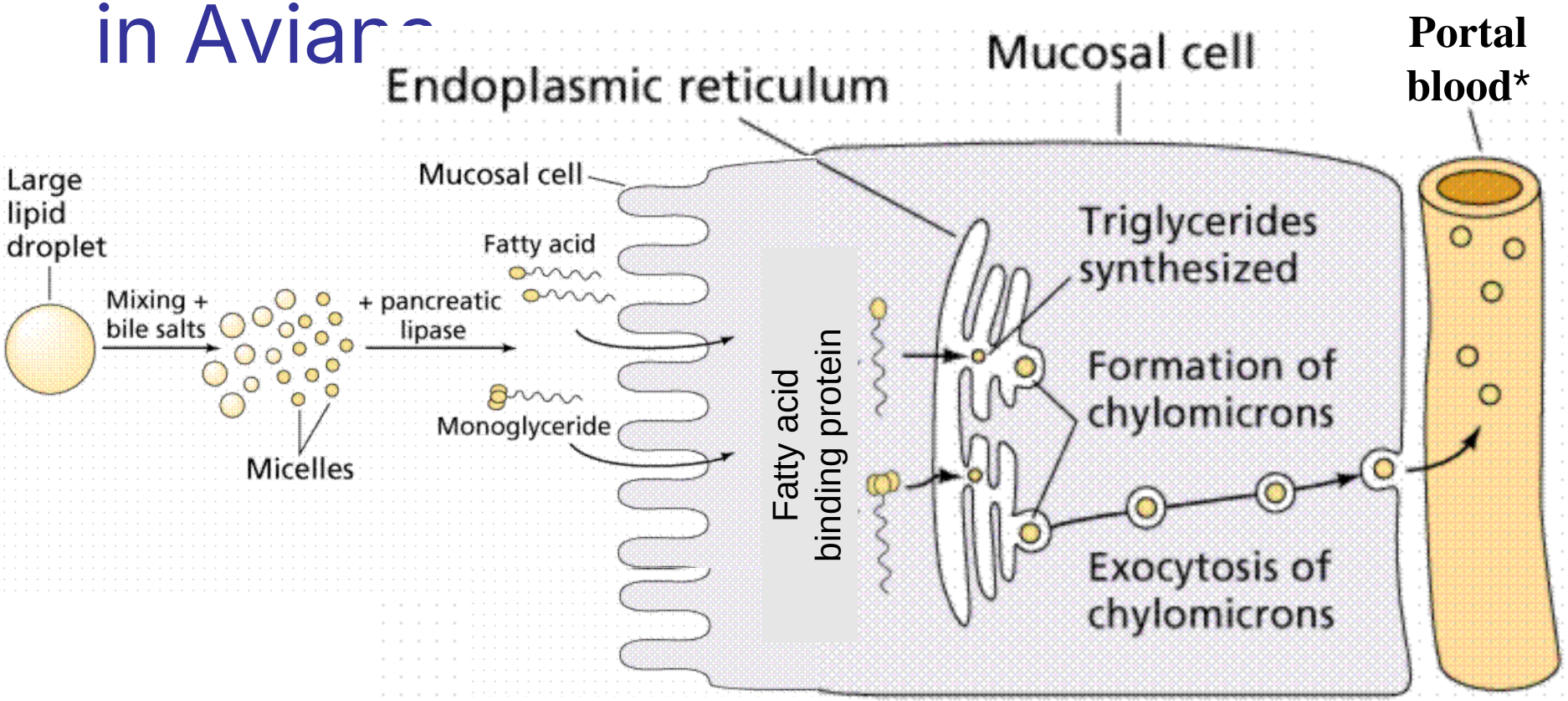
Nutrient Absorption - Lipids

- Fatty acids, 2-monoglycerides, cholesterol, and cholesterol esters move down concentration gradient (passive diffusion)
- Repackaged in intestinal cell for transport to liver
  - Some is reformed into triglycerides
  - Chylomicrons

Lipid Absorption (Chylomicrons)

- Chylomicrons absorbed from enterocytes into lacteals (lymph vessels)
  - Ultimately enter blood via thoracic duct
    - Most long chain fatty acids absorbed into lymphatic system
      - Exception is poultry
- Blood lipids transported as lipoproteins

Overview of Lipid Digestion and Absorption in Aviar



*Lymph in mammals

Lipid Digestion - Ruminants

- Microbes rapidly modify lipids:
- Lipolysis

  Triglycerides ⟶ Glycerol + 3 free fatty acids
- Biohydrogenation
  - Addition of H to unsaturated fatty acids
    - Saturation
  - If carried to completion, all double bonds become single bonds

Lipid Digestion and Synthesis by Microbes

- Rumen microbes
  - Produce “trans” configured double bonds
  - Alter chain length
  - Change position of double bonds
  - Produce odd-chain and branched-chain FA
- Rumen adipose tissue varies greatly from dietary fat
  - Dietary fat must be rumen protected to affect animal

Effect of Lipid on Rumen Fermentation

- Excess amounts of unsaturated fatty acids and triglycerides
  - Decrease methane production
  - Impair fiber digestion
  - Form soaps
  - Alter rumen metabolism towards propionate production – less acetate
    - Decrease milk fat
  - Produce trans fatty acids
    - Inhibit lipid synthesis in mammary gland
      - Decrease milk fat

Lipid Digestion - Ruminant

- Digestion and absorption of lipids is similar to monogastrics except
  - Fat enters small intestine in different form than was presented to animal in diet
  - Lipids absorbed more slowly

Clinical Case

- 8 y.o. girl
  - Admitted for heart/lung transplantation
- Medical history
  - Xanthomas at 2 yo
  - MI symptoms at 7 yo
    - TC=1240mg/dl
    - TG=350mg/dl
    - Diet & statin & cholestyramine
    - Mother TC= 355, father TC=310
  - Coronary artery bypass at 7 yo
  - 8 yo severe angina, second bypass
    - TC = 1000mg/dl
- Transplantation successful
  - TC=260mg/dl, xanthomas regressing



Plasma Lipoproteins Classes & Functions

- Chylomicrons
  - Synthesized in small intestine
  - Transport dietary lipids
  - 98% lipid, large sized, lowest density
  - Apo B-48
    - Receptor binding
  - Apo C-II
    - Lipoprotein lipase activator
  - Apo E
    - Remnant receptor binding

Plasma Lipoproteins Classes & Functions

- Very Low Density Lipoprotein (VLDL)
  - Synthesized in liver
  - Transport endogenous triglycerides
  - 90% lipid, 10% protein
  - Apo B-100
    - Receptor binding
  - Apo C-II
    - LPL activator
  - Apo E
    - Remnant receptor binding

Plasma Lipoproteins Classes & Functions

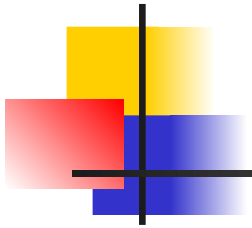
- Intermediate Density Lipoprotein (IDL)
  - Synthesized from VLDL during VLDL degradation
  - Triglyceride transport and precurser to LDL
  - Apo B-100
    - Receptor binding
  - Apo C-II
    - LPL activator
  - Apo E
    - Receptor binding

Plasma Lipoproteins Classes & Functions

- Low Density Lipoprotein (LDL)
  - Synthesized from IDL
  - Cholesterol transport
  - 78% lipid, 58% cholesterol & CE
  - Apo B-100
    - Receptor binding

Plasma Lipoproteins Classes & Functions

- High Density Lipoprotein (HDL)
  - Synthesized in liver and intestine
  - Reservoir of apoproteins
  - Reverse cholesterol transport
  - 52% protein, 48% lipid, 35% C & CE
  - Apo A
    - Activates lecithin-cholesterol acyltransferase (LCAT)
  - Apo C
    - Activates LPL
  - Apo E
    - Remnant receptor binding



- THANK YOU